Solutions to Mathematics Problems

1. Solve $e^{(x-1)} = 2e^{(3x-4)}$ correct to 4 significant figures:

Solution:

$e^{(x-1)} = 2e^{(3x-4)}$

Divide through by $e^{(x-1)}$:

$1 = 2e^{(2x-3)}$

$\ln(1/2) = 2x - 3$

$x = (3 - \ln(2)) / 2$ approximately equals 1.153 (4 significant figures).

2. Solve the equations by elimination method:

$7x - 2y = 26$

$6x + 5y = 29$

Solution:

Using elimination, $x = 4$, $y = 1$.

3. Prove the trigonometric identity:

$(1 + \cot(\theta)) / (1 + \tan(\theta)) = \cot(\theta)$

Solution:

Using $\cot(\theta) = \cos(\theta)/\sin(\theta)$ and $\tan(\theta) = \sin(\theta)/\cos(\theta)$, we simplify both sides to prove the identity.

4. Solve $\log_3(81) + \log_3(x) = 5$:

Solution:

$\log_3(81) = 4$, so $4 + \log_3(x) = 5$.

$\log_3(x) = 1$ implies $x = 3$.

5. Determine the root of $x^2 - 3x - 4 = 0$ using completing the square:

Solution:

$x^2 - 3x = 4$

$(x - 3/2)^2 = 25/4$

$x = 4$ or $x = -1$.

6. Prove the identity $(1 + \sin(\theta)) / \cos(\theta) = \cos(\theta) / (1 - \sin(\theta))$:

Solution:

Both sides simplify to $\cos^2(\theta) = 1 - \sin^2(\theta)$, which is true.

7. Solve $(2^{(x-5)})(5^{(x+1)}) = 62.5$:

Solution:

Rewrite and solve using logarithms to isolate x.

8. Solve $\log(x^2 - 3) - \log(x) = \log(2)$:

Solution:

$\log((x^2 - 3) / x) = \log(2)$

$x^2 - 3 = 2x$

$x = 3$ (valid solution).

9. Solve the simultaneous equations:

$x + y + z = 4$

$2x - 3y + 4z = 33$

$3x - 2y - 2z = 2$

Solution:

Using elimination and substitution methods to solve for x, y, and z.

10. Prove $\cot(2x) + \csc(2x) = \cot(x)$:

Solution:

Express $\cot(2x)$ and $\csc(2x)$ in terms of $\sin(2x)$ and $\cos(2x)$ and simplify to $\cot(x)$.